

FOR SMALL DEFLECTIONS THE SPRING FORCE REMAINS VERTICAL

$$\sum M_A = 0 \quad \uparrow \quad 600(2.85) - 1.5 F_s = 0$$

$$F_s = 1140 \text{ N}$$

$$F_s = k_s \delta_s \Rightarrow \delta_s = 1140 \text{ N} / 35 \times 10^3 = 32.5 \text{ mm}$$

NON RIGID SLIGHTLY LESS DEFLECTION

2.2 Consider the situation when the valve is just about to open. At this point, there is no pressure between the valve and its seat, and the fluid pressure acts on an area of 50 mm diameter.

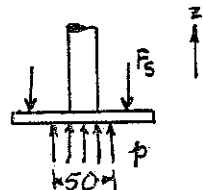
Geometry: The spring is compressed 43 mm to a length of 207 mm

Force-Deformation: Spring force  $F_s = (120 \text{ kN/m}) \times 43 \times 10^{-3} \text{ m} = 5.16 \text{ kN}$

Force Equilibrium: Consider the valve free-body

$$\sum F_z = 0: F_p = F_s, \text{ or } p \cdot \frac{\pi}{4} (50)^2 = 5.16$$

$$\therefore p = 2.63 \times 10^3 \text{ kN/m}^2$$

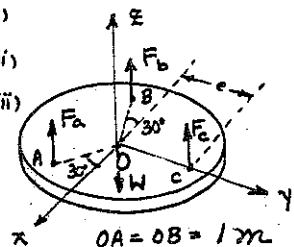


2.3 Let  $\delta_a, \delta_b, \delta_c$  be the deflections of springs a, b, c.

Geometric Compatibility:  $\delta_a = \delta_b = \delta_c$

Force Deformation:  $F_a = k_a \delta_a, k_a = 14 \text{ kN/m} \dots (i)$   
 $F_b = k_b \delta_b, k_b = 14 \text{ kN/m} \dots (ii)$   
 $F_c = k_c \delta_c, k_c = 16 \text{ kN/m} \dots (iii)$

Equilibrium of Disc: Consider the free-body:



$$\sum M_{Ox} = 0: e F_c = (F_a + F_b) \times 1 \times \sin 30^\circ$$

$$\sum F_z = 0 \quad F_a + F_b + F_c = W = 1.1 \text{ kN}$$

SOLVE

$$S = 0.025 \text{ m}$$

$$F_a = F_b = 0.35 \text{ kN}$$

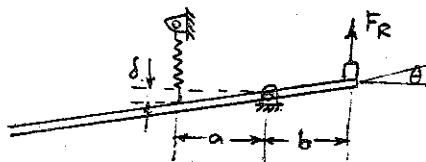
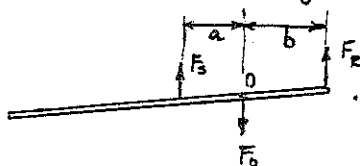
$$F_c = 0.4 \text{ kN}; e = 0.875 \text{ m}$$

2.4 When the operator removes her foot at the bottom of the stroke, the spring is extended an amount  $\delta$ , and the moment of the spring force about the pivot just balances the force in the rod.

Geometry:  $\delta \approx a\theta$  for small  $\theta$ .

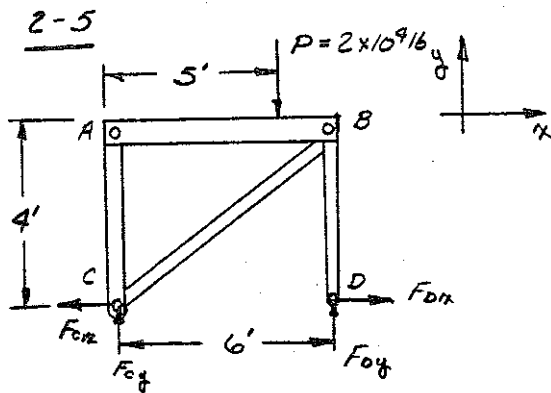
Force-Deformation:  $F_s = k\delta$

Equilibrium: consider free-body



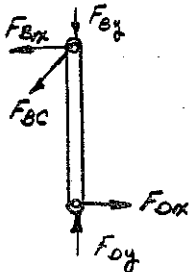
$$\sum M_O = 0: a F_s = F_R \cdot b \quad \text{or} \quad F_s = \frac{b}{a} F_R \quad \text{----- (i)}$$

We see from (i) that the moment of the spring force about O must have a constant value,  $b F_R$ , independent of the position of the spring and of the spring constant. Thus the force to be exerted by the operator cannot be altered by any alteration in the spring or its position. The only practical solution appears to be to increase the length of the lever, so that the operator's foot is at a greater distance from O. Note that this will require the operator to move her foot through a greater distance, since  $\theta$  is presumably fixed by the punch press operation. Note also that although the maximum applied force is reduced, the work done by the operator on each stroke will remain the same with the extended lever arm. It is possible that the fatigue caused will decrease.



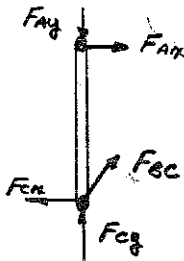
$$\begin{aligned}
 (1) \sum F_x &= 0 \\
 F_{Ax} - F_{Bx} &= 0 \\
 (2) \sum F_y &= 0 \\
 -P + F_{By} + F_{Dy} &= 0 \\
 (3) + \sum M_C &= 0 \\
 6F_{Dy} - 5P &= 0 \\
 F_{Dy} &= \frac{5}{6}P \\
 F_{Cy} &= \frac{1}{6}P
 \end{aligned}$$

FOR LINK BD



$$\begin{aligned}
 + \sum M_B &= 0 \\
 4F_{Dx} &= 0 \quad \text{AND FROM (1)} \\
 F_{Cx} &= 0
 \end{aligned}$$

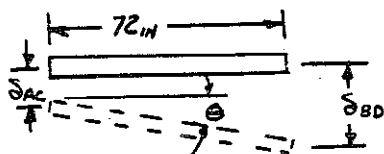
FOR LINK AC



$$\begin{aligned}
 + \sum M_C &= 0 \\
 -4F_{Ax} &= 0 \\
 \sum F_x &= 0 \\
 F_{Ax} + F_{Cx} + F_{BC} \frac{6}{13} &= 0 \\
 \text{BUT } F_{Ax} = F_{Cx} = 0 \quad \text{AND } F_{BC} &= 0
 \end{aligned}$$

$$\delta_{AC} = \frac{F_{Cy} L_{AC}}{A_{AC} E} = \frac{\frac{1}{6} P L_{AC}}{A_{AC} E} = \frac{1}{6} \frac{(2 \times 10^4 \text{ lb})(48 \text{ in})}{(15 \text{ in}^2)(30 \times 10^6 \frac{\text{lb}}{\text{in}^2})} = .3 \times 10^{-3} \text{ in}$$

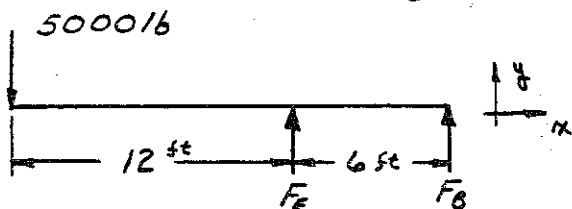
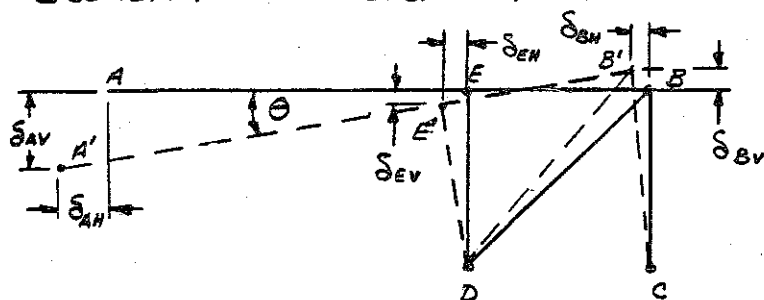
$$\delta_{BD} = \frac{F_{Dy} L_{BD}}{A_{BD} E} = \frac{\frac{5}{6} P L_{BD}}{A_{BD} E} = \frac{5}{6} \frac{(2 \times 10^4 \text{ lb})(48 \text{ in})}{(10 \text{ in}^2)(30 \times 10^6 \frac{\text{lb}}{\text{in}^2})} = 2.6 \times 10^{-3} \text{ in}$$



$$\theta = \frac{\delta_{BD} - \delta_{AC}}{72 \text{ in}} = 3.2 \times 10^{-5} \text{ RAD}$$

2-6

# GEOMETRY OF DEFORMATION



$$\sum F_y = 0$$

$$F_E + F_B - 5000 = 0$$

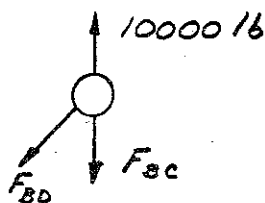
$$\sum M_E = 0$$

$$(5 \times 10^3)(12 \text{ ft}) + F_B(6 \text{ ft}) = 0$$

$$F_B = -10000 \text{ lb}$$

$$F_E = 15000 \text{ lb}$$

FOR THE PIN B:



$$\sum F_x = 0$$

$$F_{BD} = 0$$

$$\sum F_y = 0$$

$$F_{BC} = 10000 \text{ lb TENSION}$$

BY A SIMILAR ANALYSIS  $F_{DE} = 15000 \text{ COMP.}$

ROD BD DOES NOT DEFORM. IT ONLY ROTATES ABOUT D. THUS  $\delta_{BH} = \delta_{BV}$  (FROM GEOMETRY)

$$\delta_{BV} = \left( \frac{FL}{AE} \right)_{BC} = \frac{(10^4 \text{ lb})(6 \text{ ft})(12 \frac{\text{in}}{\text{ft}})}{(1 \text{ in}^2)(30 \times 10^6 \frac{\text{lb}}{\text{in}^2})} = .024 \text{ in}$$

$$\delta_{EV} = \left( \frac{FL}{AE} \right)_{ED} = \frac{(1.5 \times 10^4 \text{ lb})(6 \text{ ft})(12 \frac{\text{in}}{\text{ft}})}{(2 \text{ in}^2)(30 \times 10^6 \frac{\text{lb}}{\text{in}^2})} = .018 \text{ in}$$

FOR  $E'B'$  TO EQUAL  $EB$ ,  $\delta_{EH} \approx \delta_{BH}$  BUT  $\delta_{BH} = \delta_{BV} = .024 \text{ in}$   
 $\delta_{EH} \approx .024 \text{ in}$

SIMILARLY  $\delta_{AH} = .024 \text{ in}$

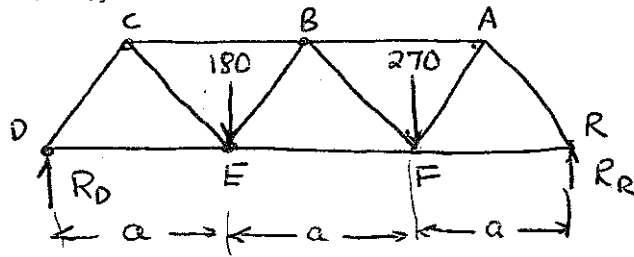
$$\theta \approx \frac{\delta_{BV} + \delta_{EV}}{12 \text{ in}} = \frac{.042}{12} \text{ RAD}$$

$$\delta_{AV} = -\delta_{BV} + \theta \times (18 \text{ ft})(12 \frac{\text{in}}{\text{ft}}) = .126 \text{ in} - .024 \text{ in}$$

$$= .102 \text{ in}$$

$$\delta_{AH} = .024 \text{ in} \quad 2-4$$

## 2.7 IDEALIZATION: TRUSS WITH FRICTIONLESS JOINTS



CALCULATE REACTIONS  
END LOADS INTO SUPPORTS

$$\sum M_D = 0 \quad \rightarrow \quad a \cdot 180 + 2a \cdot 270 = 3a R_R$$

$$R_R = 240 \text{ kN}$$

$$\sum F_y = 0 \Rightarrow R_D = 210 \text{ kN}$$

FIND FORCES IN DE, EF, FR: PIN D

$$F_{DE} + F_{DC} \cos 60^\circ = 0$$

$$F_{DC} \sin 60^\circ + R_D = 0$$

$$F_{DC} = -R_D / \sin 60^\circ$$

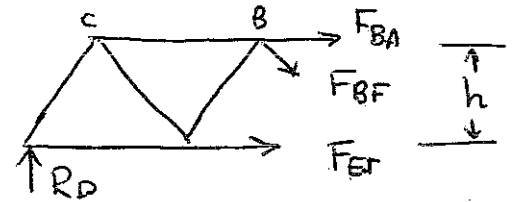
$$F_{DE} = -F_{DC} \cos 60^\circ = R_D \cot 60^\circ$$

TO FIND  $F_{EF}$  TAKE A SECTION AS SHOWN

$$\sum M_B = 0 \quad \rightarrow$$

$$180 \frac{a}{2} - R_D \frac{3a}{2} + h F_{EF} = 0$$

$$F_{EF} = 210 \times 3 \times \left(\frac{a}{2h}\right) - 180 \left(\frac{a}{2h}\right) = 450 \cot 60^\circ$$



NOW PIN R:

$$F_{FR} = R_R \cot 60^\circ$$

TO CALCULATE DEFLECTION OF POINT R

$$\delta_R \approx \left(\frac{FL}{AE}\right)_{DE} + \left(\frac{FL}{AE}\right)_{EF} + \left(\frac{FL}{AE}\right)_{FR}$$

$$\approx 3.6 \text{ mm}$$

2.8 Geometric Compatibility. Referring to Fig. 2.8, example 2.5, we see that the deflections are very small and thus bars AD, DB remain essentially in a straight line

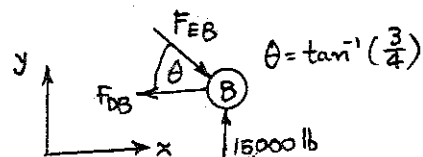
$\therefore \delta_B \approx \delta_{AD} + \delta_{DB}$ , where  $\delta_{AD}$ ,  $\delta_{DB}$  are elongations of bars AD and DB, and  $\delta_B$  is the movement of point B.

Force Equilibrium: From example 1.4,  $F_{AD} = 13,300$  lb, Tension

To obtain  $F_{DB}$ , consider a free-body of joint B:

$$\sum F_y = 0: \frac{3}{5} F_{EB} = 15,000$$

$$\therefore F_{EB} = 25,000 \text{ lb, compression}$$



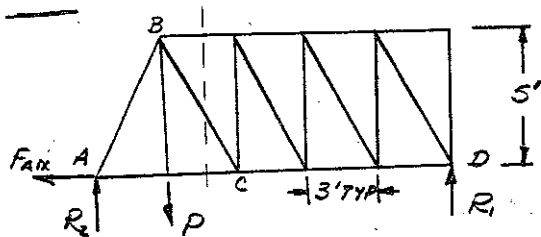
$$\sum F_x = 0: F_{DB} = \frac{4}{5} F_{EB} = 20,000 \text{ lb, tension}$$

$$\text{Force - Deformation: } \delta_{AD} = \left( \frac{Fl}{AE} \right)_{AD} = \frac{13,300 \times 8}{4 \times 10 \times 10^6} = 0.00267 \text{ ft, extension}$$

$$\delta_{DB} = \left( \frac{Fl}{AE} \right)_{DB} = \frac{20,000 \times 8}{4 \times 10 \times 10^6} = 0.004 \text{ ft, extension.}$$

$$\therefore \underline{\delta_B = 0.00267 + 0.004 = 0.00667 \text{ ft}} \quad (\text{CHECK COMPUTER SOLUTION})$$

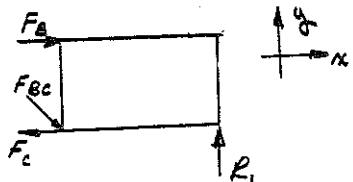
2-9



$$\sum M_A$$

$$15 R_1 - 3 P = 0$$

$$R_1 = \frac{1}{5} P = \frac{1}{5} (10 \times 10^4 \text{ lb}) = 2 \times 10^4 \text{ lb}$$



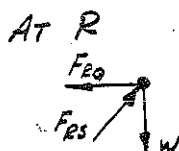
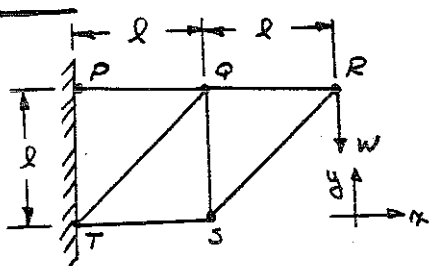
$$\sum F_y = 0$$

$$2 \times 10^4 \text{ lb} - F_{BC} \frac{5}{\sqrt{34}} = 0$$

$$F_{BC} = \frac{\sqrt{34}}{5} (2 \times 10^4 \text{ lb})$$

$$\delta_{BC} = \left( \frac{FL}{AE} \right)_{BC} = \frac{\sqrt{34}}{5} \frac{(2 \times 10^4 \text{ lb}) \sqrt{34} \text{ ft} \left( \frac{12 \text{ in}}{\text{ft}} \right)}{(2.5 \text{ in}^2) (30 \times 10^6 \frac{\text{lb}}{\text{in}^2})} = 2.1 \times 10^{-2} \text{ in}$$

2-10



$$\sum F_x = 0$$

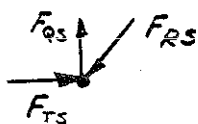
$$-F_{RQ} + \frac{F_{RS}}{\sqrt{2}} = 0$$

$$\sum F_y = 0$$

$$\frac{F_{RS}}{\sqrt{2}} - W = 0$$

$$F_{RS} = \sqrt{2} W (C), F_{RQ} = W (T)$$

At S



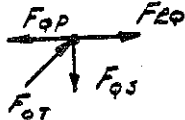
$$\sum F_x = 0$$

$$F_{TS} - \frac{F_{RS}}{\sqrt{2}} = 0, F_{TS} = W (C)$$

$$\sum F_y = 0$$

$$F_{QS} - \frac{F_{RS}}{\sqrt{2}} = 0, F_{QS} = W (T)$$

At Q



$$\sum F_y = 0$$

$$\frac{F_{QT}}{\sqrt{2}} - F_{QS} = 0, F_{QT} = \sqrt{2} W (C)$$

$$\sum F_x = 0$$

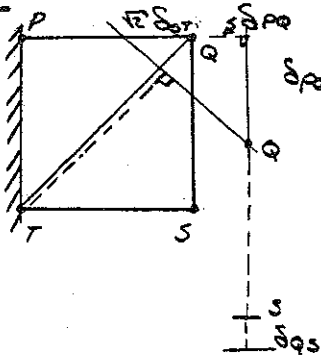
$$-F_{QP} + \frac{F_{QT}}{\sqrt{2}} + F_{RQ} = 0, F_{QP} = 2W (T)$$

$$\delta_{PQ} = \frac{F_{QP} l}{AE} = \frac{2Wl}{AE} \text{ (ELONGATION)}$$

$$\delta_{QT} = \frac{F_{QT} \sqrt{2} l}{AE} = \frac{\sqrt{2} W \sqrt{2} l}{AE} = \frac{2Wl}{AE} \text{ (COMPRESSION)}$$

$$\delta_{QS} = \frac{F_{QS} l}{AE} = \frac{Wl}{AE} \text{ (ELONGATION)}$$

2-10 (cont)



FOR THE VERTICAL DEFLECTION OF S, CONSIDER THE PORTION OF THE TRUSS SHOWN

$$\delta_{VQ} = \delta_{PQ} + \sqrt{2} \delta_{QT}$$

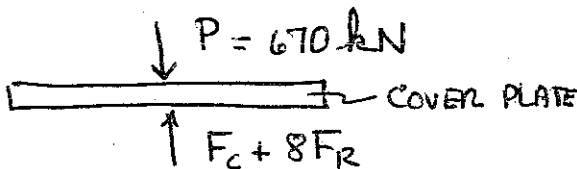
$$= (1 + \sqrt{2}) \frac{2Wl}{AE}$$

$$\delta_{VS} = \delta_{VQ} + \delta_{QS}$$

$$= (1 + \sqrt{2}) \frac{2Wl}{AE} + \frac{Wl}{AE}$$

$$= (3 + 2\sqrt{2}) \frac{Wl}{AE}$$

2-11



$$\sum F_y = 0$$

$$-P + F_C + 8F_R = 0$$

GEOMETRIC COMPATIBILITY

$$\delta_C = \delta_R$$

$$\left( \frac{PL}{AE} \right)_C = \left( \frac{PL}{AE} \right)_R$$

$$A_C = 0.3 \text{ m} \times 0.3 \text{ m}$$

$$A_R = 8 \times 25 \text{ mm} \times 25 \text{ mm}$$

$$E_C = 17 \text{ GN/m}^2$$

$$E_R = 200 \text{ GN/m}^2$$

$$\frac{F_C}{85 \times 10^{-3} \text{ m}^2 \times 17 \text{ GN/m}^2} = \frac{F_R}{625 \times 10^{-6} \times 200 \text{ GN/m}^2}$$

$$F_R = 0.0865 F_C$$

$$F_C = 395 \text{ kN}$$

$$F_R = 34.2 \text{ kN}$$

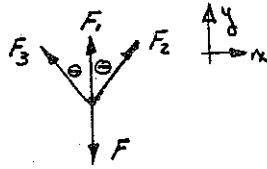
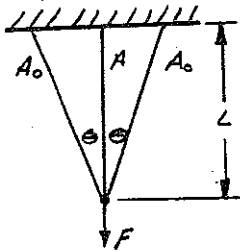
$$\sigma_C = 4.64 \text{ MN/m}^2 ; \quad \sigma_R = 54.7 \text{ MN/m}^2 , \quad \delta_R = \delta_C = \frac{\sigma_R L}{E}$$

$$= 0.328 \text{ mm}$$

2-8



2-12



$$\begin{aligned}\sum F_x &= 0 \\ -F_3 \sin \theta + F_2 \sin \theta &= 0 \\ F_3 &= F_2\end{aligned}$$

$$\begin{aligned}\sum F_y &= 0 \\ -F + F_1 + F_2 \cos \theta + F_3 \cos \theta &= 0 \\ F_1 + 2F_2 \cos \theta &= F\end{aligned}$$

GEOMETRIC COMPATIBILITY

FOR SMALL DEFLECTIONS  $\theta$  WILL REMAIN APPROXIMATELY CONSTANT



$$\delta_{A_0} = \delta_A \cos \theta$$

$$\frac{F_2 \frac{L}{\cos \theta}}{A_0 E} = \frac{F_1 L \cos \theta}{A E}$$

$$F_2 = \frac{F}{\left( \frac{A}{A_0 \cos^3 \theta} + 2 \cos \theta \right)}$$

$$F_1 = \frac{F}{\left( 1 + 2 \frac{A_0}{A} \cos^3 \theta \right)}$$

2-13

WHEN THE LOAD IS AT  $x = \lambda a$ , FROM (d)

$$\delta_A = \delta(\text{LOAD AT } x = \lambda a, x = -a) = P \frac{2k_c - \lambda(k_B + 2k_c)}{(k_A k_B + 4k_A k_C + k_B k_C)}$$

FOR THE LOAD AT  $x = -a$ , i.e.  $\lambda = -1$

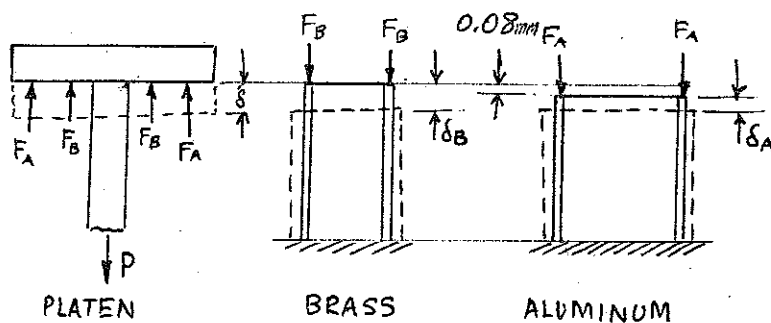
$$\delta = \delta(\text{LOAD AT } x = -a, x = \lambda a) = \delta_B + (\delta_C - \delta_B) \lambda$$

IN GENERAL

$$\frac{\delta_C - \delta_B}{P} = \frac{k_A - k_C + \lambda(k_B + k_A + k_C)}{(k_A k_B + 4k_A k_C + k_B k_C)}$$

WHEN  $\lambda = -1$

$$\delta = P \frac{2k_c - \lambda(k_B + 2k_c)}{(k_A k_B + 4k_A k_C + k_B k_C)} = \delta_A$$



Geometry: For  $0 \leq \delta \leq 0.08 \text{ mm}$ ,  $\delta_A = 0$ ,  $\delta_B = \delta$   
 For  $\delta > 0.08 \text{ mm}$ ,  $\delta_A = \delta - 0.08 \text{ mm}$ ,  $\delta_B = \delta$

FORCE DEFLECTION BRASS  $\frac{F_B}{A_B} = \frac{E_B}{L_B} \delta_B$  or  $F_B = \frac{\pi \times 150 \times 6.25 \times 10^3}{205} \delta_B$

$E_B = 103 \text{ GN/m}^2$

$E_A = 70 \text{ GN/m}^2$

ALUMINUM

$F_A = \frac{\pi \times 250 \times 6.25 \times 70}{250} \delta_A$

EQUILIBRIUM

F.B.D. OF PLATE

$P = F_A + F_B$

SOLUTION FOR  $\delta \leq 0.08 \text{ mm}$

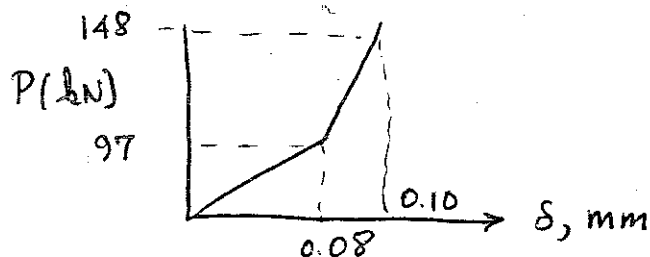
$\delta_B = \delta$ ,  $\delta_A = 0 \therefore F_A = 0$ ,  $F_B = P$

$P = 1213 \delta_B \text{ (kN)}$  ( $\delta, \text{mm}$ ) at  $\delta = 0.08$   $P = 97$

FOR  $\delta > 0.08 \text{ mm}$

$\delta_B = \delta$ ;  $\delta_A = \delta - 0.08$ ;  $P = F_A + F_B$

$P = 1374(\delta - 0.08) + 1213 \delta \text{ (kN)}$

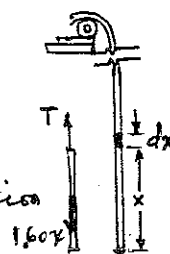


2.15

The rope will elongate under its own weight. If this elongation is less than 20 ft, a weight may be attached to the end of the rope to elongate it, if the stress in the rope does not cause it to break.

Consider a length  $x$  ft, measured from the bottom. The tension at  $x$  is

$T = 1.60x \text{ lb.}$



## 2.15 continued

The elongation of a length  $dx$  is  $\frac{1.60x}{10^6} dx$  inches  
 the total elongation of 5980 ft is  $\delta = \int_0^{5980} \frac{1.60x}{10^6} dx$  inches

or  $\delta = \frac{1.6}{2 \times 10^6} (5980)^2$  inches = 2.4 ft. This is quite inadequate.

What weight  $W$  will elongate the rope about 12 ft., so that its end is about  $5\frac{1}{2}$  ft from the floor?

$$12 = \frac{W}{10^6} \times \frac{5980}{1 \times 12} \quad \text{or} \quad \underline{W = \frac{10^6 \times 144}{5980} = 24,000 \text{ lb.}}$$

The steel rope will probably be able to bear this load and its own weight.

2.16 Idealisation: (i) Rigid spreader rigidly attached to the mast

(ii) Cables pass over spreader ends without friction.

(iii) deck is rigid

(iv) The lengths of the turnbuckles are negligible.

Geometry of Deformation Let  $L_3 \equiv AB + BC$

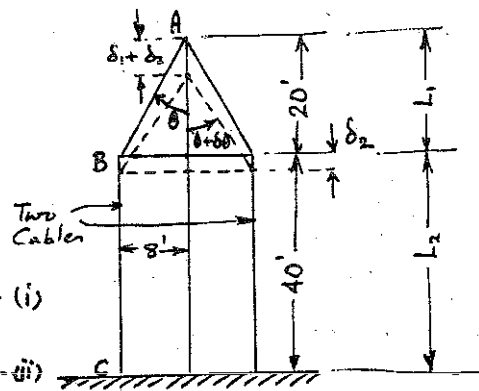
$\delta_1$  = shortening of upper portion of mast

$\delta_2$  = shortening of lower portion of mast

$\delta_3$  = shortening of cables

Undeformed:  $\sqrt{L_1^2 + 8^2} + L_2 = L_3$  ---- (i)

Deformed:  $\sqrt{(L_1 - \delta_1)^2 + 8^2} + (L_2 - \delta_2) = L_3 - \delta_3$  ---- (ii)



Also,  $\delta_3 = \delta_3' - \delta_3''$ , where  $\delta_3'$  = decrease in length of cables due to tightening of turnbuckle

$\delta_3''$  = increase in length of cables due to tension.

Equilibrium: Consider free-bodies. Since it is assumed that the ropes run over frictionless guides and are weightless, the

## 2.16 CONTINUED

tension in each rope is uniform,  $= T$ , say.  
 Since deflections are expected to be small, the changes in the various angles will be small, and we assume them to be zero in the equilibrium equations.

Upper Mast:  $F_1 = 2T \cos \theta \times 2 = 4T \cos \theta$  ---- (iii)

Lower Mast:  $\sum F_x = 0$ : satisfied

$\sum F_y = 0: 4T \cos \theta - 4T - F_1 + F_2 = 0$  ---- (iv)

From (iii) and (iv),  $T = F_2/4$ ;  $F_1 = F_2 \cos \theta$

Expanding (iii), using  $(1+\epsilon)^n \approx 1+n\epsilon$ , for  $\epsilon \ll 1.0$ ,

$$\sqrt{L_1^2 + 8^2} - \frac{S_1 L_1}{\sqrt{L_1^2 + 8^2}} + L_2 - \delta_2 = L_3 - \delta_3$$

Using (i) in this,  $-\frac{S_1 L_1}{\sqrt{L_1^2 + 8^2}} = \delta_2 - \delta_3$ , or  $-0.93 \delta_1 = \delta_2 - \delta_3$  ---- (v)

$\delta_1 = \frac{F_1 L_1}{A_1 E_1} = \frac{F_2 \cos \theta \times 20}{20 \times 1.6 \times 10^6} = 5.8 \times 10^{-7} F_2$  ft. Similarly  $\delta_2 = 12.5 \times 10^{-7} F_2$  ft.

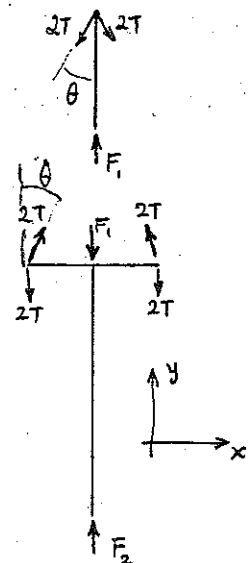
$\delta_3' = \frac{15}{20} \times \frac{1}{12} = 0.0625$  ft.

$\delta_3'' = \frac{T L_3}{12 \times 140,000} = 3.67 \times 10^{-5} T$  ft. } and  $\delta_3 = \delta_3' - \delta_3'' = 0.0625 - 9.18 \times 10^{-6} F_2$   
 $= 9.18 \times 10^{-6} F_2$  ft }

Substituting for  $\delta_3$  in (v), and using the expressions for  $\delta_1$  and  $\delta_2$ ,

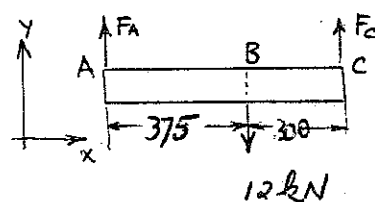
$$-(0.93)(5.8 \times 10^{-7}) F_2 = 12.5 \times 10^{-7} F_2 - 0.0625 + 9.18 \times 10^{-6} F_2$$

or  $F_2 = 5700$  lb.



2.17 Consider a free-body of the beam:  
 Assume that the moments at A, C are zero. This can be shown to be only an approximation to the true situation.

It is possible, with the theory developed in Chapters 7 & 8 to make an accurate analysis & determine the moments. Since we are



2.17 CONT.

not in a position to make such an analysis, let us make the above assumption.

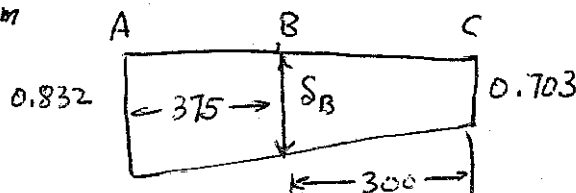
EQUILIBRIUM  $\sum M_A = 0 : F_C \times 675 = 120 \times 375 ; F_C = 66.7 \text{ kN}$

$\sum F_y = 0 : F_C + F_A = 120 : F_A = 53.3 \text{ kN}$

GEOMETRIC COMPATIBILITY  $\delta_A = \text{ELONGATION OF ROD AT A}$   
 $\delta_C = \text{ " " " " " C}$

$\delta_A = \frac{F_A l_A}{A_A E} = \frac{53.3 \times 2}{625 \times 205} = 0.832 \text{ mm}$

$\delta_C = \frac{66.7 \times 2.7}{1250 \times 205} = 0.703 \text{ mm}$

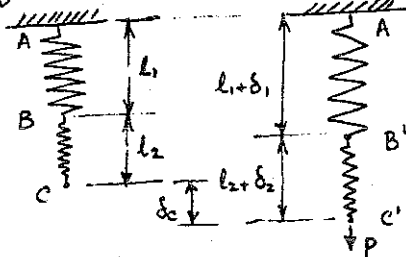


$\delta_B = 0.703 + \frac{300}{675} (0.129) = 0.760 \text{ mm}$

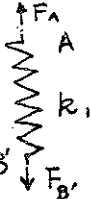
2.18

Before loading

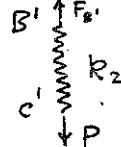
After loading



Free-Body of AB'



Free-Body of B'C'



Equilibrium:  $B'C' : F_{B'} = P$

$AB' : F_A = F_{B'} = P$

Force-Deflection:  $\delta_1 = \frac{P}{k_1} \quad \delta_2 = \frac{P}{k_2}$

Geometric Compatibility: displacement of C =  $(l_1 + \delta_1) + (l_2 + \delta_2) - (l_1 + l_2) = (\delta_1 + \delta_2)$   
 or  $\delta_c = P \left( \frac{1}{k_1} + \frac{1}{k_2} \right)$

$\therefore \text{Overall spring constant} = \frac{F}{\delta_c} = \frac{k_1 k_2}{k_1 + k_2}$

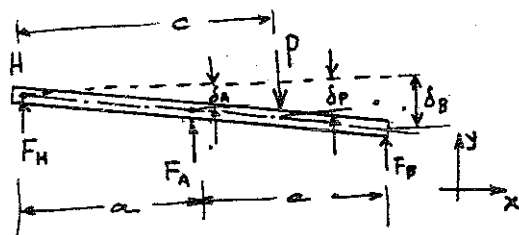
2.19

Idealisation: (i) no friction at pivots (ii) Beam is weightless (iii) Springs do not deflect sideways.

Geometry of Deflection for rigid beam:

$\delta_A = \frac{1}{2} \delta_B, \delta_P = \frac{c}{2a} \delta_B$

Force-Deformation:  $F_A = k \delta_A = \frac{k}{2} \delta_B$   
 $F_B = k \delta_B$



## 2.19 continued

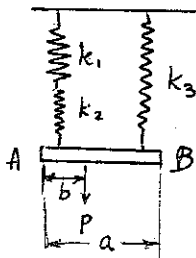
Equilibrium:  $\sum M_A = 0$ :  $P \cdot c = F_A \cdot a + F_B \cdot 2a = \frac{5}{2} k a \delta_B$

or  $P = \frac{5}{2} k \delta_B \frac{a}{c}$

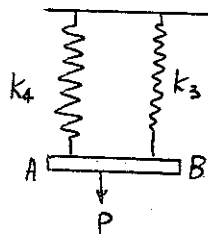
$\therefore \frac{P}{\delta_P} = \frac{\frac{5}{2} k \delta_B \frac{a}{c}}{\frac{c}{2a} \delta_B} = 5 \frac{a^2}{c^2} k$

We want  $\frac{P}{\delta_P} = \frac{20}{9} k$ .  $\therefore 5 \frac{a^2}{c^2} k = \frac{20}{9} k$ , or  $c = 1.5a$

## 2.20

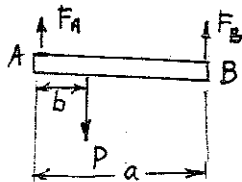


Replace this by:  
(Using the result of 2.18)



where  $k_4 = \frac{k_1 k_2}{k_1 + k_2}$

Geometry: Springs deflect by an equal amount  $\delta$ , in order that AB should remain horizontal.



Force-Deformation:

$F_A = k_4 \delta$  ----- (i)

$F_B = k_3 \delta$  ----- (ii)

Equilibrium:  $F_A + F_B = P$ ,  $F_A \cdot a = P(a-b)$

$\therefore F_A = P(1 - \frac{b}{a})$ ,  $F_B = P \frac{b}{a}$

From (i) and (ii),  $\frac{F_A}{k_4} = \frac{F_B}{k_3}$ , or  $\frac{(b-a)}{a k_4} = \frac{b}{a k_3}$

This gives  $\frac{b}{a} = \frac{k_3}{k_3 + k_4} = \frac{k_3}{k_3 + \frac{k_1 k_2}{k_1 + k_2}} = \frac{k_3 (k_1 + k_2)}{k_1 k_2 + k_3 (k_1 + k_2)}$

and  $\delta = \frac{P(k_1 + k_2)}{k_1 k_2 + k_3 (k_1 + k_2)}$

## 2.21

Geometric Compatibility: Fig (ii)

$\delta_B = \delta_A - 600 \theta$   
 $\delta_D = \delta_A - 900 \theta$   
 $\delta_C = \delta_A - 1200 \theta$

}  $\delta$ 's are in mm  
 }  $\theta$  is in radians

Force-Deflection:  $F_A = k_1 \delta_A$ ,  $F_B = k_2 \delta_B$   
 $F_C = k_3 \delta_C$

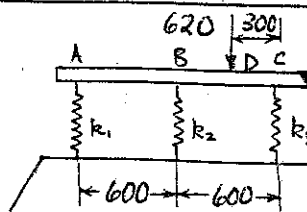


Fig (i)

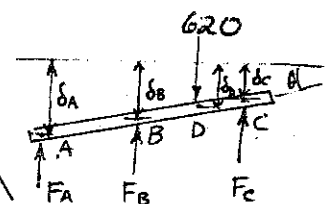


Fig (ii)

2.21 (Cont.)

THESE BECOME

$$F_A = 17 \delta_A$$

$$F_B = 21(\delta_A - 600\theta)$$

$$F_C = 17(\delta_A - 1200\theta)$$

[N],  $\delta$  [mm]

EQUILIBRIUM:

$$\sum F_y = 0 \quad F_A + F_B + F_C = 620 \text{ N}$$

$$55 \delta_A - 3.3 \times 10^4 \theta = 620$$

$$\sum M_{D2} = 0 \quad 3F_A + F_B = F_C$$

$$55 \delta_A = -7800 \theta$$

SOLVE TO FIND:

$$\theta = -1.519 \times 10^{-2} \text{ Rad} : \delta_A = 2.15 \text{ mm}$$

$$F_A = 36 \text{ N}$$

$$F_B = 236 \text{ N}$$

$$F_C = 346 \text{ N}$$

2.22 Idealisation: (i) Ball joint is frictionless, does not transmit moments

(ii) Flagpole is rigid

Symmetry: The pole deflects so as to remain in the N-S plane.

Geometry: Assume deflections are small, so that change in angles is negligible.

Deformation of ropes ① and ②:

$$AB' \approx AB + BB' \cos 45^\circ \quad BB'A \approx AB + BB' \cos 45^\circ$$

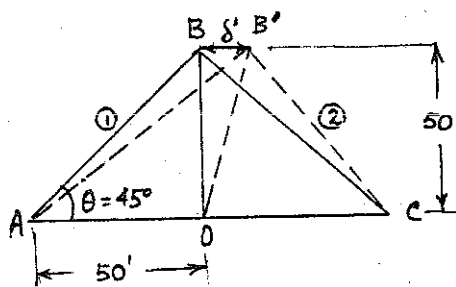
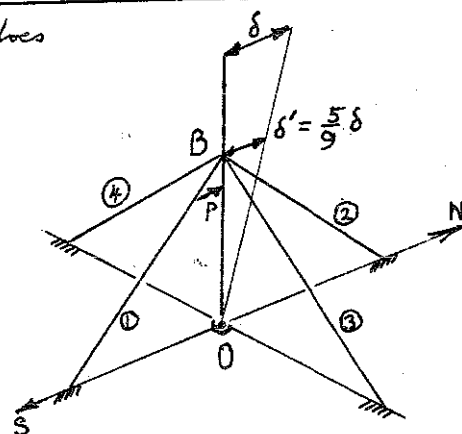
$$= AB + \delta' \cos 45^\circ$$

$$\therefore \delta_1 = \text{elongation of rope ①} = \delta' \cos 45^\circ$$

$$= \frac{5}{9} \times 0.707 \delta$$

The tendency for rope ② is to get compressed. This is not possible. ② goes slack. Because the pole moves in the plane of ① and ②, the elongation of ③ and ④ is negligible. Since the initial tensions were zero, the final tensions are zero in ②, ③ and ④.

Force-Deformation for ①:  $T_1 = A_1 E_1 \left( \frac{\delta_1}{L_1} \right) = \frac{\pi}{4} \left( \frac{1}{4} \right)^2 \times 30 \times 10^6 \times \frac{\frac{5}{9} \times 0.707 \delta}{1.414 \times 50}$



## 2.22 continued

This gives  $T_1 = 8230\delta$  where  $\delta$  is in feet,  $T$  in lb.

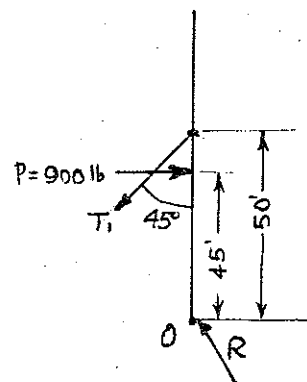
Equilibrium of pole:

$$\sum M_O = 0: T_1 \left(\frac{1}{\sqrt{2}}\right) \times 50 = 900 \times 45$$

$$\therefore T_1 = 1140 \text{ lb}$$

This gives  $8230\delta = 1140$

$$\therefore \delta = 0.139 \text{ ft} = 1.68''$$



2.23 Idealisation: pins are frictionless, rods are weightless, axial-force members. Deformations are small, so that angles change negligibly.

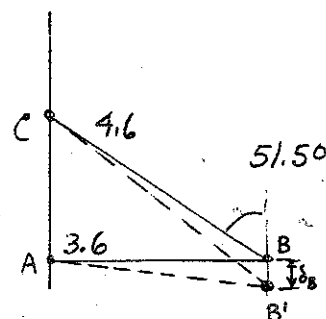
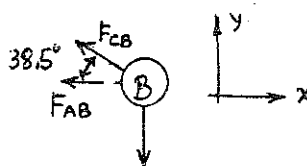
From the test in Fig(b) of the problem, the spring constant of CB is

$$k = \frac{400 \text{ N}}{23 \text{ mm}} = 17.39 \text{ N/mm}$$

Free-body of pin at B:

$$\sum F_y = 0: F_{CB} \cos 51.5^\circ = 400$$

$$\text{or } F_{CB} = 642.5 \text{ N}$$

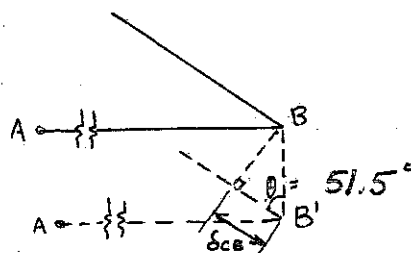


Force-Deformation:  $\delta_{CB} = \frac{F_{CB}}{k} = \frac{642.5}{17.39} = 36.95 \text{ mm}$

Geometric Compatibility: to determine  $BB' = \delta_B$  in terms of  $\delta_{CB}$ , draw an enlarged view of joint B'. In this,  $CB'$  is almost parallel to  $CB$ ,  $AB'$  is almost parallel to  $AB$ .

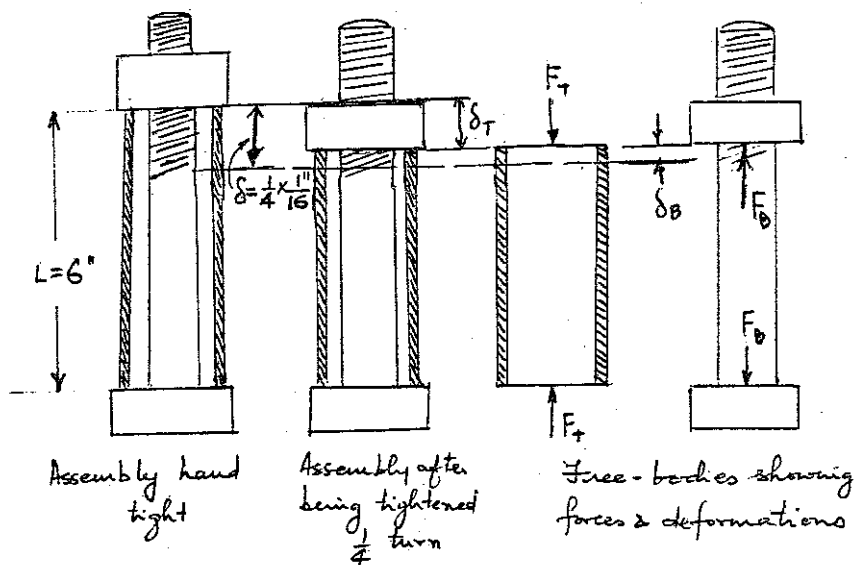
From geometry,  $BB' = \delta_B = \frac{\delta_{CB}}{\cos 51.5^\circ}$

$$\text{or } \delta_B = 59.36 \text{ mm}$$





2.24



Geometric Compatibility:

$$\delta_T + \delta_B = \delta = \frac{1}{4} \times \frac{1}{16} \text{ inch}$$

Equilibrium:  $F_T = F_B$

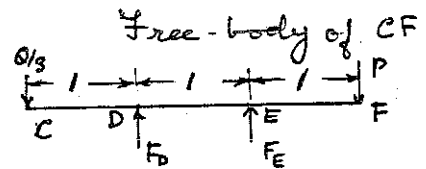
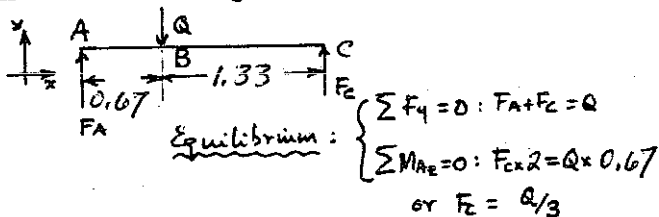
Force-Deformation:  $F_T = A_T E_T \frac{\delta_T}{L}$

$$F_B = A_B E_B \frac{\delta_B}{L}$$

$$\therefore \frac{F_T L}{A_T E_T} + \frac{F_B L}{A_B E_B} = \delta_T + \delta_B = \delta = \frac{1}{4} \times \frac{1}{16}$$

$$\text{or } \frac{F_B \times 6}{30 \times 10^6} \left( \frac{1}{0.60} + \frac{1}{1.0} \right) = \frac{1}{64} \quad \text{or } \underline{F_B = 29,400 \text{ lb}}$$

2.25 Free-body of AC



$$\sum F_y = 0: P + Q/3 = F_D + F_E \quad \dots (1)$$

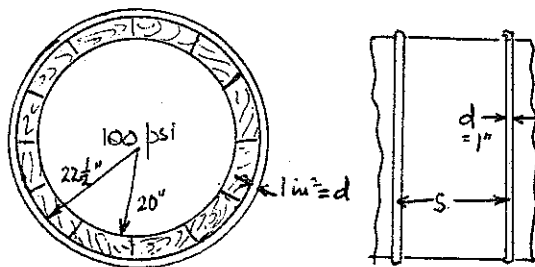
$$\sum M_{C-E} = 0: 2 F_E + 1 F_D = 3 P \quad \dots (2)$$

Geometric Compatibility: For CF to remain horizontal, deflections at D & E must be equal:  $\delta_D = \delta_E$ .

Force-Deformation:  $\delta_D = \frac{F_D L_A}{E_A A_A}$ ,  $\delta_E = \frac{F_E L_B}{E_B A_B}$ ,  $\therefore F_D = F_E \left( \frac{A_A}{A_B} \right) \left( \frac{L_B}{L_A} \right) \left( \frac{E_B}{E_A} \right) = F_E \left( \frac{1}{2} \right) \left( \frac{2}{1} \right) \left( \frac{1}{2} \right) = \frac{1}{4} F_E$

Using  $F_D = \frac{1}{4} F_E$  in (1) and (2), get  $\underline{Q = 2P}$

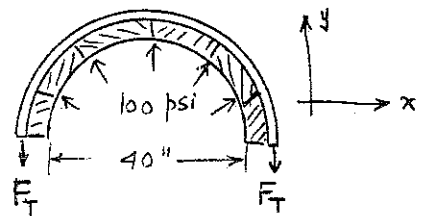
2.26



Idealization: Assume the steel hoops to be closed circles as shown.

Let the spacing between the hoops be  $S$ . The wooden staves support no tangential stress. Their only deformation is a radial compression, and is negligible. We may consider each hoop to support the pressure over a length  $S$ .

Equilibrium: Consider the free-body shown, with length  $S$  normal to the paper.



$$\sum F_y = 0: F_T \times 2 = 40 \times l \times 100 \quad \therefore F_T = 2000 S \text{ lb.}$$

$$\text{Strain (tangential)} = \frac{2000S}{\frac{\pi}{4} d^2 E} = \frac{2000S}{\frac{\pi}{4} (1)^2 30 \times 10^6} = \frac{dL}{L}, \quad L = \text{circumference of hoop}$$

$$L = 2\pi R \quad \therefore dL = 2\pi dR, \quad \frac{dL}{L} = \frac{dR}{R} = \frac{2000l}{\frac{\pi}{4} 30 \times 10^6}$$

We require  $dR \leq \left(\frac{0.03}{2}\right)$ . The limiting case is  $dR = \left(\frac{0.03}{2}\right)$ .

$$\therefore l = \frac{\pi 30 \times 10^6}{4 \times 2000} \times \frac{(0.03)^2}{2} = \underline{\underline{765''}}$$

2.27 From example 2.8, (d)

$$\frac{dT}{d\theta} = fT, \quad \text{where } T = \text{tension at angle } \theta$$

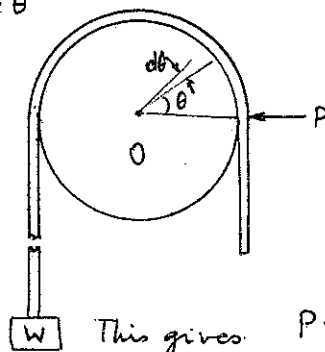
Integrating,  $T = Ce^{f\theta}$ ,  $C = \text{const.}$

$$\text{at } \theta = \pi, T = W = Ce^{f\pi} \quad \therefore C = We^{-f\pi}$$

$$\text{and } T = We^{f(\theta-\pi)}$$

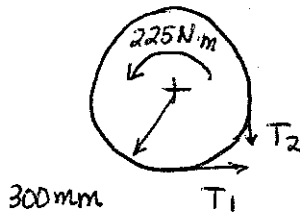
$$\text{at } \theta = 0, T_0 = We^{-f\pi}$$

In order for this to be possible, we require a force such that  $fP = T_0$ .



$$\text{This gives } P = \frac{W}{f} e^{-f\pi}$$

2-28



$$\sum M_o = 0 \rightarrow$$

$$225 \text{ N.m} - 0.3 T_2 + 0.3 T_1 = 0$$

$$\text{FROM FRICTION: } T_2 = T_1 e^{f\theta}$$

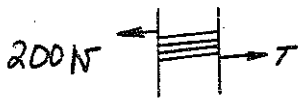
$$\theta = \frac{3}{2}\pi = 1.5\pi$$

$$T_2 = T_1 e^{0.4(1.5\pi)} = 6.586 T_1$$

$$T_1 = 134 \text{ N}$$

$$T_2 = 884 \text{ N}$$

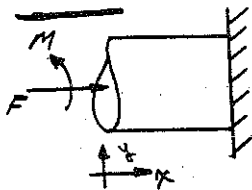
2-29



$$\frac{T}{200} = e^{f\theta} = e^{0.3(8\pi)}$$

$$T = 3.76 \times 10^5 \text{ N}$$

2-30



$$\begin{aligned} \rightarrow \sum M &= 0 \\ M - \int dM_R &= 0 \end{aligned}$$

$$\int dM_R = \int r dF_R \quad (\text{RESISTING MOMENT})$$

LET  $p$  BE A UNIFORM PRESSURE ON THE LOWER SURFACE

$$dF_R = \int dF = \int p dA = \int p r dr d\theta$$

$$\int dM_R = \int \int_0^{2\pi} \int_0^r p r^2 dr d\theta = \int \frac{2}{3} p r^3 \pi$$

$$\sum F_x = 0$$

$$F - p \pi r^2 = 0 \quad p = \frac{F}{\pi r^2}$$

$$\int F = \text{RESISTING FORCE} = \frac{3}{2} \frac{M}{r}$$

2.31

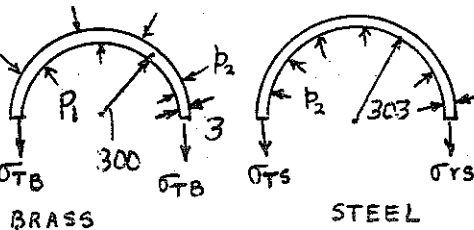
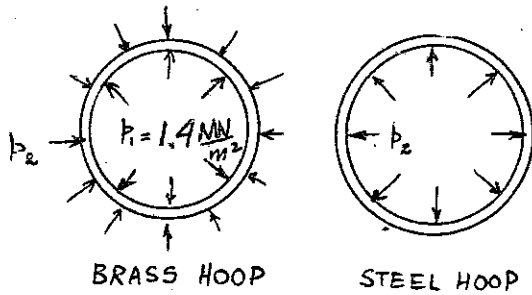
Geometric Compatibility: Symmetry requires that tangential displacements be zero for both hoops. Thus there will be no shear stresses between the two hoops, and their interaction consists of a radial pressure.

Their radial expansion must be equal:

$$(\delta_R)_{\text{Brass}} = (\delta_R)_{\text{Steel}}$$

Consider the semi-circular free-bodies, with unit width normal to the paper.

Equilibrium gives



$$\sigma_{TB} = \frac{(p_1 - p_2) \times 0.3}{3 \times 10^{-3}} \quad ; \quad (\delta_R)_B = \frac{(p_1 - p_2)(0.3)^2}{3 \times 10^{-3} \times 103 \times 10^3}$$

$$E_B = 103 \times 10^3 \text{ MN/m}^2$$

$$E_S = 205 \times 10^3 \text{ MN/m}^2$$

$$\sigma_{TB} = \frac{p_2 \times 0.303}{6 \times 10^{-3}} \quad ; \quad (\delta_R)_S = \frac{p_2 (0.303)^2}{6 \times 10^{-3} \times 205 \times 10^3}$$

$$(\delta_R)_B = (\delta_R)_S$$

$$(p_1 - p_2) \times 2.91 \times 10^{-4} = (\delta_R)_B$$

$$p_2 \times 0.7464 \times 10^{-9} = (\delta_R)_S$$

$$p_2 = 1.11 \text{ MN/m}^2$$

$$\sigma_{TS} = 56.3 \text{ MN/m}^2 \quad ; \quad F_{TS} = 20,270 \text{ N}$$

$$\sigma_{TB} = 28.6 \text{ MN/m}^2 \quad ; \quad F_{TB} = 5,144 \text{ N}$$

2.32

In problem 2.31 using (iii), (iv) and  $(\delta r)_B = (\delta r)_S$ , get.

$$p_2 \approx 1.11 \text{ MN/m}^2 \quad ; \quad (\delta r)_B = 0.083 \text{ mm}$$

2.33

From (b) of Example 2.8,  $dN = \frac{dT}{f}$  ----- (i)

and from (e),  $T = T_{AD} e^{f\theta}$  ----- (ii)

$\therefore$  Combining (i) and (ii),  $\frac{dN}{R d\theta} = \frac{1}{f} \cdot \frac{f T_{AD} e^{f\theta} d\theta}{R d\theta} = \frac{T_{AD} e^{f\theta}}{R}$ , ----- (iii)

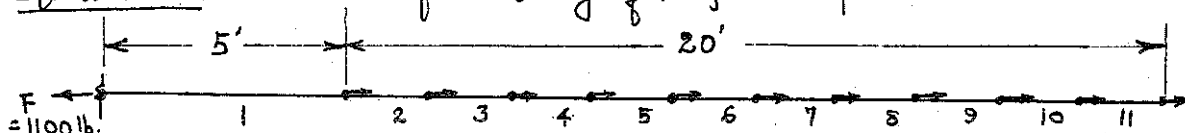
where  $dT = f T_{AD} e^{f\theta} d\theta$ .

This is the required result.

2.34

Idealisation: rope is weightless; all forces are axial.

Equilibrium: consider a free-body of half the rope



In the sections marked 1, 2, 3, ..., 10, 11, the tensile forces are 1100 lb, 1000 lb, ..., 100 lb. The lengths are 5 ft, 2 ft, 2 ft, ..., 2 ft. The cross-sectional area is the same everywhere,  $\frac{\pi}{4} (\frac{1}{2})^2 \text{ in}^2$ .

Force-deformation: For the  $j^{\text{th}}$  section,  $\delta_j = \frac{F_j L_j}{29,400}$ , where  $\delta_j = \text{extension of } j^{\text{th}} \text{ section}$ ,  $L_j = \text{length of } j^{\text{th}} \text{ section}$ .

Total elongation =  $\sum_{j=1}^{11} \delta_j = \frac{1}{29,400} [1100 \times 5 + 2 \times (1000 + 900 + \dots + 100)] = \underline{0.56 \text{ ft.}}$

$\therefore$  Length of rope when pulled =  $2 \times (25 + 0.56) = 51.12 \text{ ft.}$

Desired Clearance = 4 ft on each side = 8 ft overall.

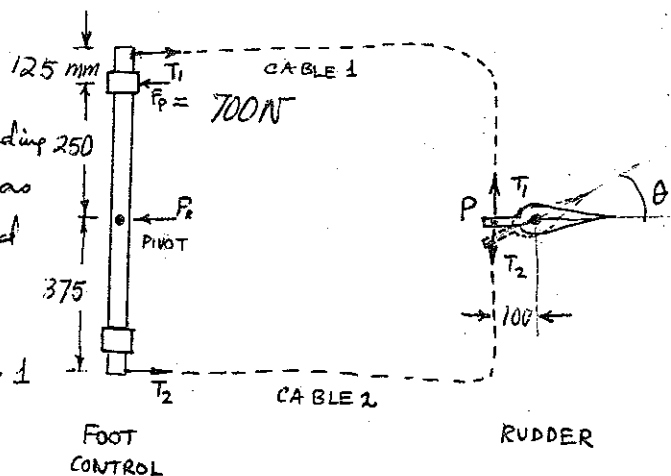
$\therefore$  LENGTH OF FIELD REQUIRED =  $51.12 + 8 = \underline{59.12 \text{ ft.}}$

2-35

Assume that the foot control does not move. Let

$L$  = length of each cable before loading

$\delta$  = distance that point P moves as a result of the force applied to the rudder.



Geometric Compatibility: cable 1

must stretch by  $\delta$ , cable 2

must shrink by  $\delta$ , or become

slack. Thus cable 2 can either be in reduced tension, or have no force at all.

Also, assuming  $\delta$  is small, the angle  $\theta$  through which the rudder must turn is  $\theta \approx \frac{\delta}{4}$

Equilibrium: Assuming pulleys are frictionless, tension in cable is transmitted unchanged along its length

Foot Control Free-Body:  $\sum M = 0$  about pivot:  $0.375 T_1 - 0.250 \times 700 - 0.375 T_2 = 0$

$$\therefore T_1 - T_2 = 466.7 \quad \text{----- (i)}$$

Force-Deformation let  $T_0$  = initial tension in cables.

$$\text{Cable 1: } \frac{\delta}{L} = \frac{T_1 - T_0}{k} ; \text{ Cable 2: } \frac{\delta}{L} = \frac{T_0 - T_2}{k}, \text{ if } T_2 > 0.$$

$$k = \frac{60 \text{ kN}}{25 \text{ m}}$$

Otherwise cable 2 goes slack and this equation yields no information.

Assume  $T_2 > 0$  and equate  $\delta$  from two equations:  $T_1 - T_0 = T_0 - T_2$

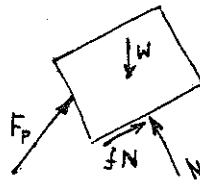
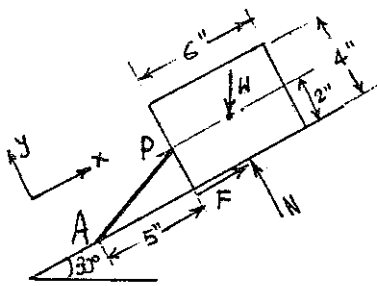
$$\text{or } T_1 + T_2 = 2T_0 = 2.8 \text{ kN} \quad \text{----- (ii)}$$

$$\begin{aligned} \text{From (i) + (ii)} \quad T_1 = 1633 \text{ N}; \quad T_2 = 1166 \text{ N} \quad \delta &= 6 \times (1633 - 1400) / (60/25) \\ &= 0.5825 \text{ mm} \quad \theta = 0.0058 \text{ rad} \\ &= 0.33^\circ \end{aligned}$$

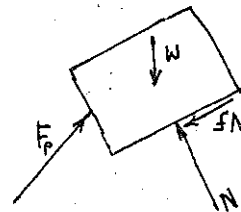
$$\begin{aligned} \text{If } T_0 = 0; \text{ ASSUME } T_2 > 0, \text{ WE GET } T_1 - T_2 &= 466.7 \quad T_1 = 233 \text{ N} \\ T_1 + T_2 &= 0 \quad T_2 = -233 \text{ N} \end{aligned}$$

$T_2$  cannot be negative. Hence we take  $T_2 = 0$  and get  $T_1 = 466.7 \text{ N}$  from (i). Thus, without initial tension the rudder would turn through an angle twice as large as it would with initial tension.

2.36



SLIDING DOWN



SLIDING UP

The case may tend to slide either up or down the plane. Which alternative is chosen depends, among other things, on the weight of the strut, the friction between the strut and the case and the manner in which the strut is constrained. We shall analyze the cases where the case is just about to slide either up or down, thus obtaining limiting values for the compression of the strut. Assume that the strut is weightless, that its contact with the case gives an axial force  $F_p$ , and that A is a pin joint. The force  $F_p$  then acts along the axis of the strut.

$$\text{Constitutive law: } -fN \leq F \leq fN$$

The extreme values of  $F$  are  $F = \pm fN$ .

$$\begin{aligned} \text{Equilibrium: } \sum F_x = 0 \quad \frac{F_p \cdot 5}{\sqrt{29}} &= W \sin 30^\circ \pm fN \\ \sum F_y = 0 \quad W \cos 30^\circ &= N + \frac{2}{\sqrt{29}} F_p \end{aligned} \quad \begin{cases} W = 10T \\ f = 0.2 \end{cases}$$

From these, the extreme values of  $F_p$  are 7650 lb and 15,400 lb.

$$\text{or } 7650 \leq F_p \leq 15,400.$$

We have  $P = 7650$  lb when the case is about to slide up,  
 $P = 15400$  lb when the case is about to slide down.

$$\text{For the shortening of the strut, } \delta = \frac{PL}{AE} = \frac{P \cdot 12\sqrt{29}}{4 \times 1.6 \times 10^6}$$

$$\therefore \text{Sliding up, } \delta = \frac{7650 \times 12\sqrt{29}}{4 \times 1.6 \times 10^6} = 0.077''$$

$$\text{Sliding down, } \delta = \frac{15400 \times 12\sqrt{29}}{4 \times 1.6 \times 10^6} = 0.135'' \quad \therefore \underline{0.077'' \leq \delta \leq 0.135''}$$

2-37

Idealization: Jaws are rigid; loads are transmitted from bolts to jaws at first thread; clearances are large enough to allow slight tilt.

Geometry: Let C be turned  $N$  turns

$\delta_B$  = elongation of bolt B

$\delta_C$  = shortening of bolt C.

Then  $a = (1 + \frac{1}{2} \delta_B)$  in, because  $a \approx \frac{1}{2} \times \text{length of bolt B}$

$$b = (1 + \frac{N}{20} - \delta_C) \text{ in.}$$

If jaws are rigid, compatibility requires  $b + 0.97 = 2a$

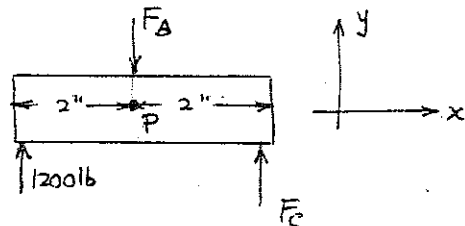
Substituting for  $a$  and  $b$ ,  $\delta_B + \delta_C = \frac{N}{20} - 0.03$  ---- (i)

Equilibrium:

$$\sum F_y = 0: 1200 + F_C = F_B$$

$$\sum M_P = 0: 1200 \times 2 = F_C \times 2$$

$$\therefore F_B = 2400 \text{ lb}, F_C = 1200 \text{ lb.}$$



Force-Deformation:  $\delta_B = \frac{F_B L_B}{A_B E_B} = \frac{2400 \times 2}{0.1 \times 30 \times 10^6} = 0.0016 \text{ in.}$

$$\delta_C = \frac{F_C L_C}{A_C E_C} = \frac{1200 \times 1}{0.1 \times 30 \times 10^6} = 0.0004 \text{ in.}$$

Substituting  $\delta_B, \delta_C$  in (i),  $N = 0.64$  turns.

Let  $\delta_{RC}$  = increase in radius of copper coil

$\delta_{RS}$  = increase in radius of steel coil.

Geometric Compatibility:  $\delta_{RS} = \delta_{RC} - 0.5 \times 10^{-3}$  ---- (i)

THE TOTAL OUTWARD FORCE =  $\pi \times 4.8 \times 70 \text{ kN}$

WE MAY REPLACE THIS BODY FORCE BY AN EQUIVALENT

INTERNAL PRESSURE

$$P_0 = \frac{\pi \times 4.8 \times 70}{\pi \times 4.8 \times 0.075} = 0.93 \text{ MN/m}^2$$

2-38



2-38 CONT

Let THE INTERACTION BETWEEN THE COPPER + STEEL BE A RADIAL PRESSURE  $p_c$ . THEN PROCEEDING EXACTLY AS IN 2.31

$$\delta_{RC} = \frac{(p_o - p_c)(2.4)^2}{0.075 \times 117 \times 10^3}$$

$$\delta_{RS} = \frac{p_c (2.4)^2}{0.025 \times 200 \times 10^3}$$

SUBSTITUTING IN (c)

$$p_c = 0.0608 \text{ MN/m}^2$$

FROM Eq (d),  
EXAMPLE (c)

$$\text{FORCE} = (p_o - p_c) 4.8 \times 0.075 = 0.313 \text{ MN}$$

2-39

Geometric compatibility:

If  $\delta$  = deflection of yoke,

$\delta_B$  = elongation of B =  $\delta$

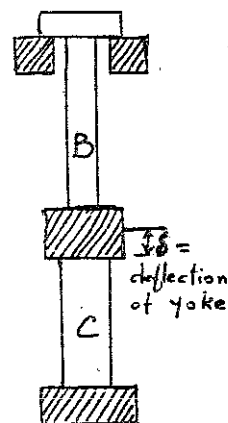
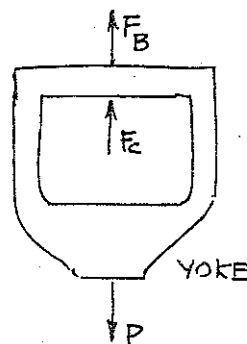
$\delta_c$  = shortening of C =  $\delta$

Equilibrium of Yoke:  $P = F_B + F_c$  --- (i)

$$(P = 325 \text{ lb})$$

Constitutive laws:

$$\frac{\delta_B}{L_B} = f_B\left(\frac{F_B}{A_B}\right), \quad \frac{\delta_c}{L_c} = f_c\left(\frac{F_c}{A_c}\right)$$



where  $f_B, f_c$  are the stress-strain curves for Bar B and C.

$$\text{using } \delta_B = \delta_c = \delta, \quad L_B f_B\left(\frac{F_B}{A_B}\right) = L_c f_c\left(\frac{F_c}{A_c}\right) \text{ --- (ii)}$$

(i) and (ii) must be solved graphically for  $F_B$  and  $F_c$ .

Procedure: we may write  $\left(\frac{F_B}{A_B}\right)F_B + \left(\frac{F_c}{A_c}\right)A_c = \left(\frac{P}{A}\right)_B \times 0.01 + \left(\frac{P}{A}\right)_c \times 0.02 = P$

$$\text{and } \left(\frac{\delta}{L}\right)_B L_B = \left(\frac{\delta}{L}\right)_c L_c \quad \therefore \left(\frac{\delta}{L}\right)_c / \left(\frac{\delta}{L}\right)_B = \frac{L_B}{L_c} = \frac{3}{2}$$

Tabulation:

| TRIAL | $(\delta/L)_B$ | $(\delta/L)_c$ | $(P/A)_B$ | $(P/A)_c$ | $F_B$ | $F_c$ | P   |
|-------|----------------|----------------|-----------|-----------|-------|-------|-----|
| 1     | 0.003          | 0.0045         | 7,500     | 13,400    | 75    | 268   | 343 |
| 2     | 0.0028         | 0.0042         | 7,000     | 13,900    | 70    | 260   | 330 |
| 3     | 0.0026         | 0.0039         | 6,500     | 12,800    | 65    | 256   | 321 |

$\approx 325$

APPROXIMATE SOLUTION

$$F_B \approx 69 \text{ lb}$$

$$F_c \approx 256 \text{ lb}$$

FROM EXAMPLE 2.2

$$(a) \sum F_y = 0 = F_c + F_o - W$$

$$(b) \sum M_c = 0 = 2a F_o - (a+b) W$$

$$(c) \frac{h_c}{h_o} = \frac{L+a}{L-a}$$

$$(d) \begin{aligned} \delta_c &= h - h_c \\ \delta_o &= h - h_o \end{aligned}$$

$$(e) \begin{aligned} F_c &= k_c \delta_c \\ F_o &= k_o \delta_o \end{aligned}$$

From (d) & (e) 
$$\begin{aligned} F_c &= k_c (h - h_c) & \text{or} & \frac{F_c - k_c h}{F_o - k_o h} = \frac{k_c h_c}{k_o h_o} \\ F_o &= k_o (h - h_o) \end{aligned}$$

WITH (c)

$$\frac{F_c - k_c h}{F_o - k_o h} = \frac{k_c}{k_o} \frac{L+a}{L-a} \quad \text{LET } r = \frac{k_c}{k_o}, \quad r = \frac{L+a}{L-a}$$

$$F_c - r F_o = k_c h - k_o h r = k_c h (1-r)$$

WITH (a)

$$F_c + F_o = W$$

$$F_o = \frac{W}{1+r} - \frac{k_c h (1-r)}{1+r} = \frac{(a+b) W}{2a} \quad \text{FROM (b)}$$

SOLVING FOR b

$$b = - \frac{(1-r)}{1+r} a \left\{ \frac{2k_c h}{W} - \frac{1-r}{1-r} \right\}$$

INSERTING THE DEFINITION OF  $r$  AND DIVIDING BY  $L$ , EQUATION (i) CAN BE OBTAINED WITH MANIPULATION

$$(i) \quad \frac{b}{L} = \frac{(a/L)^2}{1 - \frac{1}{2}(1-r)(1+\frac{a}{L})} \left\{ \frac{2k_c h}{W} - 1 + \frac{1}{2}(1-r)(1+\frac{a}{L}) \right\}$$

USING THE METHODS OF EXAMPLE 2.9 TO CALCULATE THE DEFLECTION IS VERY TIME CONSUMING. A GRAPHICAL METHOD, THE WILLIOT DIAGRAM, IS AVAILABLE (SEE FOR DISCUSSION TIMOSHENKO & YOUNG, THEORY OF STRUCTURES, MCGRAW-HILL). THE SOLUTION OF THIS PROBLEM IS BEST SUITED TO DIGITAL COMPUTATION AND IS GIVEN IN SECTION 2.5.

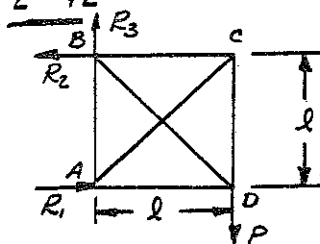
ENERGY METHOD : APPLY A UNIT LOAD AT F. THEN CALCULATE THE LOAD IN EACH MEMBER DUE TO THIS UNIT LOAD. THE RESULTS ARE GIVEN BELOW.

| i | $F_i$<br>$10^3 \text{ lb}$ | $(L/AE)^*$<br>$10^{-6} \text{ in/lb}$ | $\frac{\partial F_i}{\partial P}$ | $(\frac{FL}{AE}) \frac{\partial F_i}{\partial P} \times 10^{-3}$ |
|---|----------------------------|---------------------------------------|-----------------------------------|------------------------------------------------------------------|
| 1 | + 13.33                    | 2.4                                   | $\frac{2}{3}$                     | 21.3                                                             |
| 2 | + 20.0                     | 2.4                                   | $\frac{2}{3}$                     | 32.0                                                             |
| 3 | - 25.                      | 1.5                                   | $-\frac{5}{6}$                    | 31.3                                                             |
| 4 | - 16.67                    | 1.5                                   | $-\frac{5}{6}$                    | 20.8                                                             |
| 5 | - 16.67                    | 1.5                                   | $-\frac{5}{6}$                    | 20.8                                                             |
| 6 | - 16.67                    | 1.5                                   | $-\frac{5}{6}$                    | 20.8                                                             |
| 7 | 0                          | 1.5                                   | 0                                 | 0                                                                |
| 8 | - 8.33                     | 3.0                                   | 0                                 | 0                                                                |
| 9 | + 5.0                      | 3.6                                   | 0                                 | 0                                                                |

\*  $E = 10 \times 10^6 \frac{\text{lb}}{\text{in}^2}$

$\Sigma = .147 \text{ in}$   
 $= \delta_{F_u}$

2-42



$$\sum M_B = 0$$

$$R_1 l - Pl = 0, R_1 = P$$

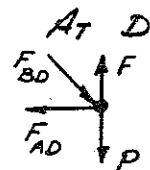
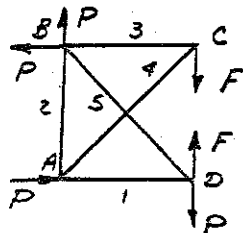
$$\sum F_x = 0$$

$$R_1 - R_2 = 0, R_2 = P$$

$$\sum F_y = 0$$

$$R_3 - P = 0, R_3 = P$$

REMOVE LINK CD AND REPLACE BY UNKNOWN FORCES, F



$$\sum F_y = 0$$

$$-\frac{F_{BD}}{\sqrt{2}} + F - P = 0, F_{BD} = \sqrt{2}(F - P)$$

$$\sum F_x = 0$$

$$-F_{AD} + \frac{F_{BD}}{\sqrt{2}} = 0, F_{AD} = (F - P)$$

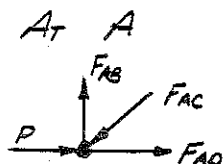


$$\sum F_y = 0$$

$$\frac{F_{AC}}{\sqrt{2}} - F = 0, F_{AC} = \sqrt{2}F$$

$$\sum F_x = 0$$

$$-F_{CB} + \frac{F_{AC}}{\sqrt{2}} = 0, F_{CB} = F$$



$$\sum F_y = 0$$

$$F_{AB} - \frac{F_{AC}}{\sqrt{2}} = 0, F_{AB} = F$$

$$U_{\text{STRUCTURE}} = \sum_i \frac{F_i^2 l_i}{2A_i E_i}$$

$$= \frac{1}{2AE} \left\{ (F-P)^2 l + F^2 l + F^2 l + (\sqrt{2}F)^2 \sqrt{2}l + [\sqrt{2}(F-P)]^2 \sqrt{2}l \right\}$$

$$U_{CO} = \frac{F_{CO}^2 l}{2AE} = \frac{F^2 l}{2AE}$$

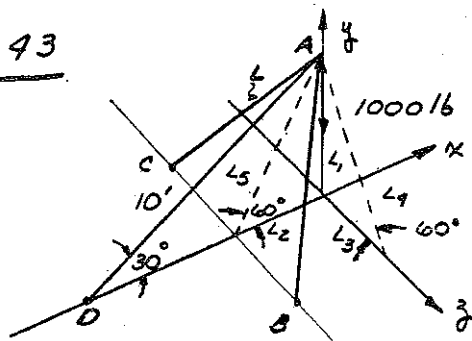
$$U_{\text{TOTAL}} = U_{CO} + U_{\text{STRUCTURE}}$$

SINCE F IS AN INTERNAL FORCE  $\frac{\partial U_{\text{TOTAL}}}{\partial F} = 0$

$$\frac{\partial U_{\text{STRUCTURE}}}{\partial F} = -\frac{\partial U_{CO}}{\partial F}$$

$$F = \frac{1 + 2\sqrt{2}}{4 + 4\sqrt{2}} P$$

2-43



$$L_1 = L_5 \sin 60 = L_9 \sin 60 = 10 \sin 30$$

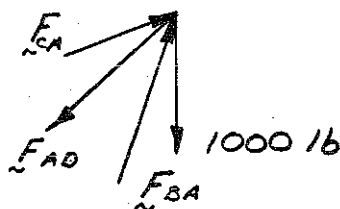
$$L_2 = L_3 = L_4 \cos 60$$

$$= 10 \frac{\sin 30}{\sin 60} \cos 60$$

$$L = [L_1^2 + L_2^2 + L_3^2]^{1/2} = 10 [(\frac{1}{2})^2 + (\frac{\sqrt{3}}{6})^2 + (\frac{\sqrt{3}}{6})^2]^{1/2} = 10 \frac{\sqrt{13}}{6}$$

$$\underline{t}_{BA} = (+L_2 \underline{i} + L_1 \underline{j} - L_3 \underline{k}) / L = (\sqrt{3} \underline{i} + 3 \underline{j} - \sqrt{3} \underline{k}) / \sqrt{13}$$

$$\underline{t}_{CA} = (+L_2 \underline{i} + L_1 \underline{j} + L_3 \underline{k}) / L = (\sqrt{3} \underline{i} + 3 \underline{j} + \sqrt{3} \underline{k}) / \sqrt{13}$$



ASSUME THE BARS DO NOT BEND AND ARE ONLY TWO FORCE MEMBERS.

$$\underline{F}_{AD} = F_{AD} (-\frac{\sqrt{3}}{2} \underline{i} - \frac{1}{2} \underline{j})$$

$$\underline{F}_{BA} = F_{BA} \underline{t}_{BA}$$

$$\underline{F}_{CA} = F_{CA} \underline{t}_{CA}$$

$$\underline{W} = -1000 \underline{j}$$

$$\Sigma \underline{F} = 0$$

$$\underline{i} \quad F_{BA} \frac{\sqrt{3}}{\sqrt{13}} + F_{CA} \frac{\sqrt{3}}{\sqrt{13}} - F_{AD} \frac{\sqrt{3}}{2} = 0$$

$$\underline{j} \quad F_{BA} \frac{3}{\sqrt{13}} + F_{CA} \frac{3}{\sqrt{13}} - F_{AD} \frac{1}{2} - 1000 = 0$$

$$\underline{k} \quad -F_{BA} \frac{\sqrt{3}}{\sqrt{13}} + F_{CA} \frac{\sqrt{3}}{\sqrt{13}} = 0$$

$$F_{BA} = F_{CA} = \frac{\sqrt{13}}{4} (1000) \text{ (COMPRESSION)}$$

$$F_{AD} = 1000 \text{ (TENSION)}$$

AT A FOR A UNIT LOAD  $\underline{P} = -\underline{j}$

$$F_{CA} = F_{BA} = \frac{\sqrt{13}}{4} \text{ (COMP)}, F_{AD} = 1 \text{ (TENSION)}$$

FOR A UNIT LOAD  $\underline{Q} = \underline{i}$

$$F_{CA} = \frac{\sqrt{3}}{2} \text{ (TENSION)}, F_{BA} = \frac{\sqrt{3}}{2} \text{ (COMP)}, F_{AD} = 0$$

## 2-43 (CON'T)

FOR A UNIT LOAD  $\bar{L} = \bar{z}$

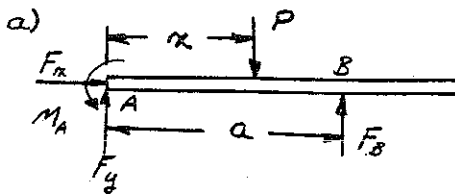
$$F_{BA} = F_{CA} = \frac{\sqrt{3}}{4} \text{ (COMP)}, \quad F_{AD} = \sqrt{3} \text{ (TENSION)}$$

|    | $F$<br>$\times 10^3 \text{ lb}$ | $L/AE$<br>$\text{IN}/16 \times 10^{-6}$ | $\frac{\partial F}{\partial P}$ | $\frac{\partial F}{\partial Q}$ | $\frac{\partial F}{\partial R}$ |
|----|---------------------------------|-----------------------------------------|---------------------------------|---------------------------------|---------------------------------|
| AD | 1.0                             | 12                                      | 1                               | 0                               | $\frac{\sqrt{3}}{4}$            |
| BA | -.97                            | 1.2                                     | $-\frac{\sqrt{3}}{4}$           | $-\frac{\sqrt{3}}{2}$           | $-\frac{\sqrt{3}}{4}$           |
| CA | -.97                            | 1.2                                     | $-\frac{\sqrt{3}}{4}$           | $\frac{\sqrt{3}}{2}$            | $-\frac{\sqrt{3}}{4}$           |

|    | $\frac{FL}{AE} \frac{\partial F}{\partial P}$<br>IN | $\frac{FL}{AE} \frac{\partial F}{\partial Q}$<br>IN | $\frac{FL}{AE} \frac{\partial F}{\partial R}$<br>IN |
|----|-----------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|
| AD | $12 \times 10^{-3}$                                 | 0                                                   | $28 \times 10^{-3}$                                 |
| BA | $1.1 \times 10^{-3}$                                | $+1.3 \times 10^{-3}$                               | $.7 \times 10^{-3}$                                 |
| CA | $1.1 \times 10^{-3}$                                | $-1.3 \times 10^{-3}$                               | $.7 \times 10^{-3}$                                 |

$$\begin{aligned} \Sigma &= .014 \text{ IN} & \Sigma &= 0 & \Sigma &= .029 \text{ IN} \\ &= \delta_y & &= \delta_z & &= \delta_x \end{aligned}$$

## 2-44



ASSUME SMALL DEFLECTIONS, THEN  
THE FORCE  $F_B$  IS VERTICAL

$$\begin{aligned} + \Sigma M_A &= 0 \\ M_A + F_B a - P x &= 0 \end{aligned}$$

$$M_A = k_3 \Theta$$

$$\delta_B = \Theta a$$

$$\delta_B = \delta_1 + \delta_2 = \frac{F_B}{k_1} + \frac{F_B}{k_2} = F_B \left( \frac{k_1 + k_2}{k_1 k_2} \right)$$

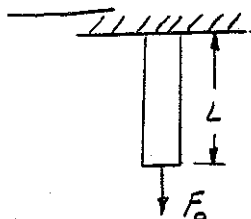
$$\Theta = \frac{F_B}{a} \left( \frac{k_1 + k_2}{k_1 k_2} \right)$$

$$\begin{aligned} k_3 \frac{F_B}{a} \left( \frac{k_1 + k_2}{k_1 k_2} \right) + F_B a &= P x \\ F_B &= \frac{P x}{\frac{k_3}{a} \left( \frac{k_1 + k_2}{k_1 k_2} \right) + a} \end{aligned}$$

b)

$$\Theta = \frac{P x}{k_3 \left( \frac{k_1 + k_2}{k_1 k_2} \right) + a^2} \left( \frac{k_1 + k_2}{k_1 k_2} \right)$$

2-45



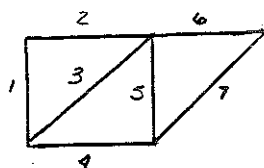
$$U = \frac{F_0^2 L}{2AE}$$

For A BAR WEIGHT DENSITY OF  $w = \frac{16}{IN}$

$$U = \frac{F_0^2 L}{2AE} + \frac{F_0 w L}{2AE} + \frac{w^2 L}{6AE}$$

$$\sigma_x = \left( \frac{F_0 + w x}{A} \right)$$

2-46



ANALYSIS OF THE TRUSS YIELDS  
THE FOLLOWING RESULTS

| DESIGN | MAX. TENSILE<br>FORCE | MAX. TENSILE<br>STRESS | MAX COMP.<br>LOAD | COMPRESSIVE<br>PROPERTIES                    | $P_c < P_{cr}$ | WT. |
|--------|-----------------------|------------------------|-------------------|----------------------------------------------|----------------|-----|
| 1      | 8000                  | 3690                   | 8485              | $P_{cr} = 12450$<br>$A = 2.76 \quad I = .91$ | YES            | 318 |
| 2      | 8485                  | 3840                   | 8000              | $P_{cr} = 16100$<br>$A = 2.17 \quad I = .59$ | YES            | 314 |
| 3      | 5656                  | 2560                   | 8246              | $P_{cr} = 23800$<br>$A = 2.17 \quad I = .59$ | YES            | 272 |

MEMBER WEIGHTS

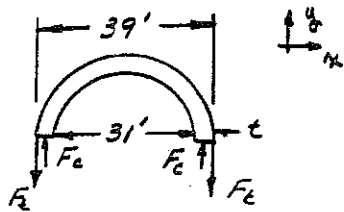
|   | 1    | 2    | 3    |   | 1    | 2    | 3    |
|---|------|------|------|---|------|------|------|
| 1 | 37.5 | 37.5 | 37.5 | 5 | 37.5 | 37.5 | 31.0 |
| 2 | 37.5 | 37.5 | 37.5 | 6 | 37.5 | 37.5 | 37.5 |
| 3 | 67.2 | 63.5 | 42.4 | 7 | 63.5 | 63.5 | 54.0 |
| 4 | 37.5 | 37.5 | 31.0 |   |      |      |      |

ALL THREE DESIGNS MEET THE STRESS CRITERIA.  
DESIGN 3 MUST BE SELECTED ON THE CONSIDERATION  
OF MINIMUM WEIGHT.

2-47

ASSUME THE TENDON FAILS AS THE CONCRETE GOES INTO TENSION

FOR PRETENSION



$$\sum F_y = 0$$

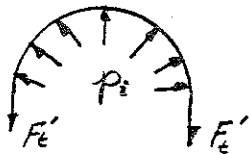
$$-2 F_t + 2 F_c = 0, F_t = F_c = F$$

STRESSES ARE GIVEN BY

$$\sigma_t = \frac{F}{\frac{\pi b^2}{4}} (T), \sigma_c = \frac{-F}{t b} (C)$$

$$\text{WHERE } \frac{\pi b^2}{4} = 170 \frac{\pi (\frac{1}{4} \text{ in})^2}{4}, b = 3.26 \text{ in} = .27 \text{ ft}$$

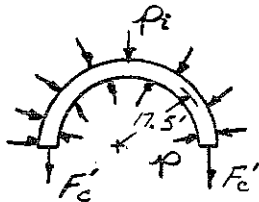
FOR INTERNAL PRESSURIZATION, ASSUME THE CONCRETE IS IN TENSION



$$\sum F_y = 0 \quad -2 F'_t + 2 r_c b p_i = 0$$

$$F'_t = p_i r_c b$$

$$\sigma'_t = \frac{F'_t}{\frac{\pi b^2}{4}} = \frac{4 p_i r_c}{\pi b} (T)$$



$$\sum F_y = 0$$

$$-2 F'_c + (p - p_i) r_c b = 0$$

$$F'_c = (p - p_i) r_c b$$

$$\sigma'_c = \frac{F'_c}{t b} = \frac{(p - p_i) r_c}{t} (T)$$

FOR GEOMETRIC COMPATIBILITY  $\delta_t = \delta_c$

$$\frac{(p - p_i) r_c^2}{t E_c} = \frac{p_i r_c^2}{b E_c}, p - p_i = p_i \frac{t E_c}{b E_c} \left( \frac{r_c}{r_e} \right)^2$$

$$p = p_i \left[ 1 + \frac{(4 \text{ ft}) (1.5 \times 10^6 \frac{\text{lb}}{\text{in}^2}) (19.5 \text{ ft})^2}{(.27 \text{ ft}) (30 \times 10^6 \frac{\text{lb}}{\text{in}^2}) (17.5 \text{ ft})^2} \right] = 1.92 p_i$$

FOR THE CONCRETE TO GO INTO TENSION  $\sigma_c + \sigma'_c = 0$ ,

$$\frac{F}{t b} = p \frac{(1 - 1.92) r_c}{t}, F = .48 p r_c b$$

FOR THE TENDON TO FAIL  $\sigma_t + \sigma'_t = 1.5 \times 10^5 \frac{\text{lb}}{\text{in}^2}$

$$\frac{F}{\frac{\pi b^2}{4}} + \frac{p_i r_c}{\frac{\pi b}{4}} = 1.5 \times 10^5 \frac{\text{lb}}{\text{in}^2}$$

$$p (.48 r_c + .52 r_e) = (1.5 \times 10^5 \frac{\text{lb}}{\text{in}^2}) \frac{\pi b}{4}, p = 1.7 \times 10^3 \frac{\text{lb}}{\text{in}^2}$$

PRETENSILE FORCE,  $F = .48 r_c b = 5.6 \times 10^5 \text{ lb}$

CONCRETE PRESTRESS  $\sigma_c = \frac{F}{t b} = 3.6 \times 10^3 \frac{\text{lb}}{\text{in}^2} < 6 \times 10^3 \frac{\text{lb}}{\text{in}^2}$

2-32



2-48

ASSUME THE LENGTH REMAINS CONSTANT AND  $t/D \ll 1$

$$V = \frac{\pi D^2}{4} L, \quad V' = \frac{\pi (D + \Delta D)^2}{4} L = \frac{\pi (D^2 + 2D\Delta D + \Delta D^2)}{4} L$$

$$\Delta V = V' - V = \frac{2\pi (D\Delta D)}{4} L \quad \text{FOR } \Delta D \ll 1$$

$$\frac{\Delta V}{V} = \frac{2\Delta D}{D}$$

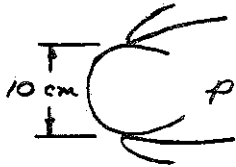
FROM THE THIN RING SOLUTION  $\sigma_r = \frac{Pr^2}{tE}$

$$\Delta D = 2\delta_r = \frac{2\Delta P (\frac{D}{2})^2}{tE}$$

$$\frac{\Delta V}{V} = \frac{2}{D} \left( \frac{2\Delta P D^2}{4tE} \right)$$

$$B = \frac{\frac{\Delta V}{V}}{\frac{\Delta P}{E}} = \frac{D}{tE} = \frac{1.25 \text{ mm}}{12.5 \text{ mm } E} = \frac{10}{E}$$

2-49



$p = 60 \text{ mm Hg}$

$$\begin{aligned} F &= \pi r^2 p \\ &= \pi (5 \text{ cm})^2 (60 \text{ mm Hg}) (1.3 \times 10^2 \frac{\text{N}}{\text{m}^2 \cdot \text{mm Hg}}) \\ &= 61.26 \text{ N} \end{aligned}$$

2-50 (a)

FOR THE STAINLESS STEEL WIRE

$$\sum F_y = 0$$

$$2F_T = 2pr \times 1.25$$

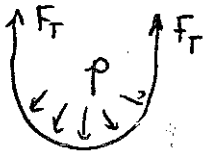
$$p = 14.4 \text{ MN/m}^2$$

FOR THE TUBE  $t/2r \ll 1$

$$\sigma_r = \frac{pr^2}{tE}$$

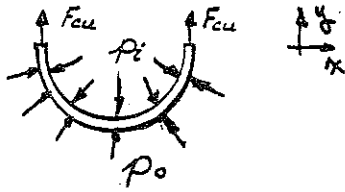
$$= \frac{14.4 \frac{\text{MN}}{\text{m}^2} (22)^2}{(2.5) (140 \times 10^3) \text{ MN/m}^2}$$

$$= 1.99 \times 10^{-2} \text{ mm}$$



## 2-50 (CON'T)

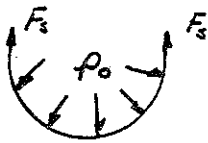
### b) PRESSURIZING THE TUBE



$$F_{cu} = p_i r_i b - p_o r_o b$$

OR IN TERMS OF THE STRESS

$$\sigma_{cu} = \frac{F_{cu}}{t b} = (p_i r_i - p_o r_o) / t$$



$$F_s = p_o r_o b, \quad \sigma_s = \frac{p_o r_o}{b}$$

FROM COMPATIBILITY  $\delta_{cu} = \delta_s$

$$\frac{(p_i - p_o) r_i}{t E_{cu}} = \frac{p_o r_o}{b E_s}$$

$$p_i = 2.76 p_o$$

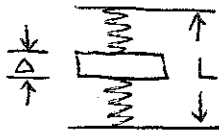
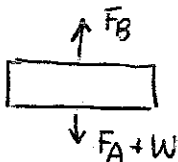
ASSUME TUBE FAILS  $\sigma_{cu} = \sigma_f = 210 \text{ MN/m}^2 = (p_i r_i - p_o r_o) / t$

$$p_o = 14.7 \text{ MN/m}^2$$

$$\sigma_s = \frac{p_o r_o}{b} = 293 \text{ MN/m}^2 < \sigma_{s \max}$$

$$p_i = 2.76 p_o = 40.4 \text{ MN/m}^2$$

2-51



$$F_A + W = F_B$$

$$k_A (L_A - L_{Ai}) + W = k_B (L_B - L_{Bi})$$

$$L_A + L_B = L_1$$

$$k_A (L_A - L_{A0} + (L_{A0} - L_{Ai})) + W = k_B (L_B - L_{B0} + (L_{B0} - L_{Bi}))$$

BUT  $k_A (L_{A0} - L_{Ai}) = k_B (L_{B0} - L_{Bi})$

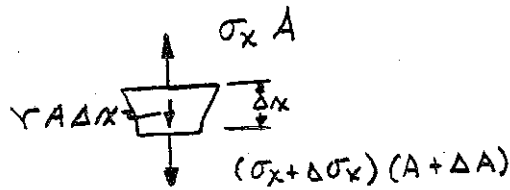
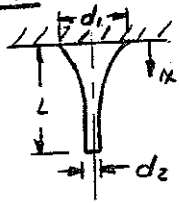
$$\therefore k_A \delta_A + W = k_B \delta_B \quad ; \quad \delta_A = L_A - L_{A0} ; \quad \delta_B = L_B - L_{B0}$$

$$(L_A - L_{A0}) + (L_B - L_{B0}) = L_1 - (L_{A0} + L_{B0}) = L_2 \equiv 0$$

$$\delta_A + \delta_B = 0$$

$$\therefore \frac{W}{\delta_B} = (k_B + k_A) \quad \text{INDEPENDENT OF } L \quad !$$

2-52



$$\sum F_x = 0 \quad -\sigma_x A + \gamma A \Delta x + (\sigma_x + \Delta \sigma_x)(A + \Delta A) = 0$$

EXPANDING, NEGLECTING PRODUCTS OF  $\Delta$  TERMS

$$\sigma_x \Delta A + A \Delta \sigma_x = -\gamma A \Delta x$$

DIVIDING BY  $\Delta x$  AND SETTING  $\sigma_x = \sigma_0$ , A CONSTANT,

$$\text{IN THE } \lim_{\Delta x \rightarrow 0} \quad \frac{dA}{dx} = -\frac{\gamma}{\sigma_0} A$$

$$\int_{A_1}^{A_2} \frac{dA}{A} = -\int_0^L \frac{\gamma}{\sigma_0} dx$$

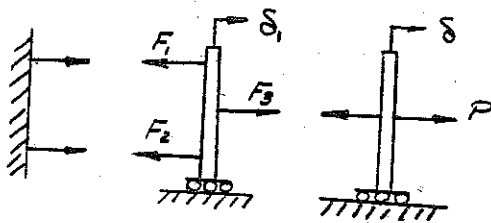
$$\ln \frac{A_2}{A_1} = -\frac{\gamma}{\sigma_0} L$$

$$2 \ln \frac{d_2}{d_1} = \frac{\gamma}{\sigma_0} L$$

$$d_1 = d_2 e^{+\frac{\gamma L}{2\sigma_0}}$$

2-53

$$\sum \int_0^{s_i} P_i \cdot ds_i = U$$



EQUILIBRIUM:  $P = F_3 = F_1 + F_2$

GEOMETRIC COMPATIBILITY:  $\delta = \delta_1 + e_3 = e_1 + e_3$   
 $e_1 = e_3$

$$\begin{aligned} \sum_{i=1}^3 \int_0^{e_i} F_i de_i &= \int F_1 de_1 + \int F_2 de_2 + \int F_3 de_3 \\ &= \int F_1 de_1 + \int F_2 de_2 + \int P d(\delta - e_1) \\ &= \int (F_1 - P) de_1 + \int F_2 de_2 + \int P d\delta \\ &= \int (F_1 + F_2 - P) de_1 + \int P d\delta \\ &= \int P d\delta \end{aligned}$$

WHICH SHOWS THAT THE TOTAL WORK DONE BY THE EXTERNAL LOAD  $P$  IS EQUAL TO THE TOTAL POTENTIAL ENERGY STORED BY THE INTERNAL MEMBERS.

2-54

PROCEEDING AS IN 2.53

$$\begin{aligned} \sum \int_0^{F_i} e_i dF_i &= \int e_1 dF_1 + \int e_2 dF_2 + \int e_3 dF_3 \\ &= \int e_1 dP + \int e_3 dP \\ &= \int \delta dP \end{aligned}$$

2-36

2-55

FOR FIG. 2.18

$$\sum \int_0^{s_i} P_i \cdot ds_i = \sum \int_0^{s_i} f_i ds_i = U$$

CONSIDER A SMALL INCREMENT IN DISPLACEMENT  $s_i$   
EQUAL TO  $\Delta s_i$ , THIS GIVES RISE TO AN  
INCREMENT IN  $U$  EQUAL TO  $\Delta U$

$$\int_0^{s_i + \Delta s_i} f_i ds_i = U + \Delta U$$

THUS

$$f_i \Delta s_i = \Delta U$$

OR

$$f_i = \frac{\partial U}{\partial s_i}$$